RS-MoE: A Vision–Language Model With Mixture of Experts for Remote Sensing Image Captioning and Visual Question Answering

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***Abstract*— Remote sensing image captioning (RSIC) presents unique challenges and plays a critical role in applications such as environmental monitoring, urban planning, and disaster management. Traditional RSIC methods often struggle to produce rich and diverse descriptions. Recently, with significant advancements in vision–language models (VLMs), efforts have emerged to integrate these models into the remote sensing domain and to introduce richly descriptive datasets specifically designed to enhance VLM training. However, most current RSIC models generally apply only fine-tuning to these datasets without developing models tailored to the unique characteristics of remote sensing imagery. This article proposes RS-MoE, the first mixture of expert (MoE)-based VLM specifically customized for remote sensing domain. Unlike traditional MoE models, the core of RS-MoE is the MoE block, which incorporates a novel instruction router and multiple lightweight large language models (LLMs) as expert models. The instruction router is designed to generate specific prompts tailored for each corresponding LLM, guiding them to focus on distinct aspects of the RSIC task. This design not only allows each expert LLM to concentrate on a specific subset of the task, thereby enhancing the specificity and accuracy of the generated captions, but also improves the scalability of the model by facilitating parallel processing of subtasks. In addition, we present a two-stage training strategy for tuning our RS-MoE model to prevent performance degradation due to sparsity. We fine-tuned our model on the RSICap dataset using our proposed training strategy. Experimental results on the RSICap dataset, along with evaluations on other traditional datasets where no additional fine-tuning was applied, demonstrate that our model achieves state-of-the-art performance in generating precise and contextually relevant captions. Notably, our RS-MoE-1B variant achieves performance comparable to 13B VLMs, demonstrating the efficiency of our model design. Moreover, our model demonstrates promising generalization capabilities by consistently achieving state-of-the-art performance on the remote sensing visual question answering (RSVQA) task.**

***Index Terms*— Mixture of experts (MoE), multimodal large language model (MLLM), remote sensing, vision–language model (VLM).**

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I. Introduction

IMAGE captioning integrates computer vision and natural language processing to automatically generate descriptive text for images, effectively bridging the information gap between visual and linguistic domains. Unlike captioning of common natural images, remote sensing image captioning (RSIC) poses greater challenges due to the inherent characteristics of remote sensing images, which cover larger geographic areas and more diverse geographical objects [1], [2]. Consequently, RSIC tasks demand not only the description of geographic object information but also a comprehensive understanding of relationships and interactions among various geographic objects.

Early studies of RSIC focused on traditional methodologies, including template-based [3] and retrieval-based methods [4], [5]. However, these methods cannot generate rich and varied descriptive sentences. Recent studies [6], [7], [8], [9], [10], [11] commonly adopt the encoder–decoder architecture, which divides the RSIC task into an image encoding phase that extracts semantic features from the input image, and a sequence modeling phase that uses the extracted features to generate text and sentences. Depending on the specific models employed, encoder–decoder approaches can be further categorized into CNN-based encoders with RNN/LSTM decoders [7] and CNN-based encoders with Transformer decoders [8]. Although these methods have achieved satisfying performance on the RSIC task, they typically generate only one or two simple sentences for captioning, limiting their practical application. This limitation typically arises due to two principal reasons: the simplicity and repetitiveness of the sentences in the datasets used for training models and the relatively

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limited ability of these models to extract semantic features and generate complex descriptions.

The large language models (LLMs) and vision–language models (VLMs) have recently achieved significant success across multiple fields, including computer vision [12], natural language processing [13], and robotics [14], [15]. In particular, VLMs [16], [17], [18] have effectively narrowed the gap between visual images and textual natural language by advancing the understanding of intermodal relationships, reaching a level of visual comprehension comparable to human capabilities. Therefore, some researchers [19], [20], [21], [22], [23], [24], [25], [26] have recently shifted their focus to applying VLMs to remote sensing visual interpretation tasks. Specifically, for RSIC tasks, the RSGPT [20] introduced the human-annotated RSICap dataset, which provides high-quality and detailed descriptions for remote sensing images. The remote sensing captions in this dataset include summarized scene descriptions, object information (color, shape, and count), and object relationships (relative position). However, RSGPT merely fine-tunes an existing VLM model on this dataset without proposing a novel model to explore and utilize the provided detailed descriptions deeply.

In this article, we propose a novel VLM based on the mixture of experts (MoE) framework, named RS-MoE, specifically customized for the RSIC task. To the best of our knowledge, this is the first work that applies the MoE framework to VLMs in the remote sensing domain. We extend the core principles of MoE to specifically address the unique challenges of remote sensing images. By decomposing the RSIC task into specialized subtasks, we leverage different expert models to facilitate more precise and contextually aware caption generation. Specifically, RS-MoE consists of three key components: the image encoder, the VLM encoder, and the MoE block. Unlike conventional MoE modules, our MoE block consists of *a novel instruction router* and *multiple lightweight LLMs*. The instruction router is designed to generate tailored prompts for each LLM. In our experiments, these prompts focus on theme comprehension, object recognition, and relationship inference. The instruction router dynamically adjusts prompts based on visual features and task instructions, offering a more precise level of control and enabling the model to generate descriptions specifically tailored to each captioning objective. Furthermore, unlike traditional MoE models that typically use feedforward networks (FFNs) as experts, we incorporate lightweight LLMs as expert models. This design allows each LLM to process complex, task-specific linguistic information efficiently and generate detailed, contextually aware descriptions while ensuring computational feasibility. In addition to the RS-MoE model, we propose a *two-stage training strategy* to fine-tune the MoE framework for the RSIC task, addressing potential model degradation caused by the sparsity that typically arises when applying MoE to remote sensing images. This sparsity arises from the lack of representation of remote sensing-specific features in models pretrained on natural images, leading to inefficiencies in capturing the complex characteristics of remote sensing imagery, which ultimately affects model performance. Moreover, to significantly reduce the number of trainable parameters, we incorporate the low-rank adaptation (LoRA) training strategy into the RS-MoE tuning stages. After fine-tuning our model on the RSICap dataset using our proposed training strategy, we evaluated it on the RSICap dataset and directly tested it on traditional datasets, such as UCM-Captions, Sydney-Captions, and RSICD, without any additional fine-tuning on these datasets. Experimental results demonstrate that our RS-MoE-7B model achieves state-of-the-art performance, while the RS-MoE-1B variant achieves performance comparable to 13B VLMs in generating precise and contextually relevant captions. Furthermore, we extend our model to the remote sensing visual question answering (RSVQA) task and demonstrate promising generalization capabilities by consistently achieving state-of-the-art performance on this task. Our contributions are summarized as follows.

1) We are the first to introduce the MoE framework to multimodal remote sensing, leveraging its task decomposition concept with specialized expert models to effectively address the complexity and diversity of remote sensing data across both visual and textual modalities.
2) We propose RS-MoE, a novel VLM specifically designed for RSIC, with an instruction router that dynamically generates task-specific prompts and lightweight LLMs as expert models to enhance both effectiveness and efficiency.
3) We present a two-stage training strategy for RS-MoE, incorporating proper initialization to mitigate sparsity-induced degradation and employing LoRA to reduce trainable parameters, improving both efficiency and manageability of the model during the training process.
4) Extensive experiments demonstrate that, with fine-tuning on a single dataset ∼3000 images, our model achieves state-of-the-art performance on five RSIC datasets and strong generalization on two RSVQA datasets. Notably, the lightweight RS-MoE-1B matches the performance of larger 13B VLMs while being significantly more efficient.

II. Related Work

A. Remote Sensing Image Captioning

RSIC poses a challenging task within the fields of computer vision and specialized linguistic modeling, aiming to generate descriptive narratives for remote sensing images by focusing on their visual attributes. Early methodologies predominantly employed template-based [3] and retrieval-based techniques [4], [5]. Template-based strategies involve creating fixed sentence structures and populating them with relevant information, such as attributes and categories displayed within the image. Conversely, retrieval-based strategies develop large-scale databases containing images and their corresponding textual descriptions. These strategies retrieve the most similar image from the database and provide the corresponding description.

Recent studies [6], [7], [8], [11] have proposed methods based on the encoder–decoder framework for achieving

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RSIC tasks. This architecture typically incorporates an image encoder to extract semantic features from images, which are then interpreted by a sequential model to produce textual annotations. Depending on the model used for the encoder, existing encoder–decoder methodologies for RSIC can be categorized into two types: CNN-based image encoders with RNN-based text decoders and CNN-based image encoders with Transformer-based text decoders. For instance, Shen et al. [6] first fine-tuned the CNN alongside the VAE in the first phase, followed by employing both Transformer and reinforcement learning to process spatial and semantic features for text generation. To incorporate the domain-specific knowledge of remote sensing, Zhao et al. [7] introduced a fine-grained and structured attention-based model that capitalizes on the structural attributes of semantic content in high-resolution RS images. Besides, Zia et al. [8] first implemented a multiscale visual feature encoder to extract detailed information from RS images and then engaged an adaptive multihead attention decoder to refine the text generation process based on the extracted multiscale features.

More recently, some researchers have begun to explore the use of VLM for the RSIC task. Wei et al. [11] introduced a vision–language aligning network that utilizes CLIP for visual feature extraction and a pretrained generative language model, GPT-2, to generate relevant descriptions for given remote sensing images. In addition, Hu et al. [20] have developed a high-quality, human-annotated RSIC dataset with detailed and rich descriptions specifically for the RSIC task, as well as an evaluation dataset for assessing general VLMs. They further fine-tuned the Instruct-BLIP model on this dataset, establishing a benchmark for both RSIC and RSVQA tasks.

B. VLM in Remote Sensing

VLMs [16], [17], [18] are advanced frameworks that integrate visual and textual modalities, enabling the interpretation and generation of visual content with corresponding linguistic descriptions. These models have achieved remarkable progress in tasks involving natural images, particularly in generating dialogs and interpreting visual data. By combining visual features with contextual cues, VLMs offer an interactive and coherent understanding of visual inputs [27], [28].

Inspired by the rapid advancements in computer vision, researchers have increasingly focused on integrating VLMs into remote sensing tasks. One of the early comprehensive reviews [29], highlighted key VLM applications within the remote sensing domain, identifying both the current challenges and future directions for research. Since then, several VLM-based models have emerged to address these challenges [19], [20], [21], [22], [23], [24], [25], [30]. For example, Zhang et al. [23] introduced EarthGPT, a model specifically designed for multisensor remote sensing image interpretation. This model integrates three core innovations: vision-enhanced perception, cross-modal understanding, and a unified instruction tuning mechanism, making it particularly effective for handling complex remote sensing data. In addition to these novel approaches, many remote sensing VLMs build upon the foundational architecture of LLaVA, adapting it to better address the specific challenges of remote sensing. RS-LLaVA enhances the standard LLaVA framework by incorporating a LoRA mechanism, optimizing the model's ability to process the multiscale and high-resolution nature of remote sensing images while maintaining computational efficiency. Similarly, H2RSVLM [22] extends LLaVA by integrating a pretrained vision encoder and LLM, connected through a multilayer perceptron (MLP) projector. This model is particularly designed to improve its handling of complex queries and mitigate incorrect generations, especially in remote sensing contexts where "hallucinations" are a common issue. GeoChat [21], also based on LLaVA, incorporates task-specific prompts and spatial capabilities, enabling it to handle multiple remote sensing tasks more effectively. These enhancements allow the model to better understand visual prompts and generate grounded object descriptions, providing more detailed and context-specific interactions in response to image- and region-level queries.

C. Mixture of Experts

The MoE framework, first introduced in [31], is a foundational technique in machine learning designed to dynamically allocate computational resources by selectively activating specialized experts for specific tasks. This approach has been extensively studied across diverse domains, including recommender systems, natural language processing, computer vision, and multimodal learning. For example, in recommender systems, the multigate MoE (MMoE) [32] was developed as a multitask learning model that explicitly models task relationships by sharing expert submodels across tasks, with separate gating networks optimizing individual tasks. Similarly, in natural language processing, the stable transferable MoE (ST-MoE) [33] enhances the practicality and robustness of sparse models, achieving state-of-the-art performance on a wide range of natural language benchmarks. In computer vision, the vision MoE (V-MoE) [34] introduced a sparsely activated Transformer model capable of scaling to 15 billion parameters, setting a new benchmark for efficiency and scalability in vision tasks. In the multimodal domain, the language–image MoE (LIMoE) [35] integrates vision and language modalities through contrastive training, enabling expert layers to dynamically partition data based on modalities. Furthermore, the MoE-LLaVA [36] activates only the top-k experts during inference, optimizing computational efficiency for large VLMs.

In contrast, the application of MoE in remote sensing remains largely underexplored. For instance, the MV-MoE [37] leverages a multitask learning framework by incorporating attention mechanisms to represent multimodal images across various scenes adaptively. Similarly, the SW-MoE [38] employs an MoE structure to activate experts based on different image scales and scenes, addressing challenges such as complex backgrounds and imbalanced datasets. Despite these advancements, existing studies are confined to single-modality applications. To the best of our knowledge, no prior work has explored the use of MoE in remote sensing for multimodal tasks. In this work, we address this gap by proposing a novel

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MoE-based VLM specifically designed to tackle the unique challenges of RSIC and VQA.

III. Methods

A. Problem Statement

Let $I \in \mathbb{R}^{W\times H\times 3}$ be a remote sensing image, where W and H denote the width and height of the image, respectively, and 3 represents the RGB color channels. The objective of RSIC is to generate a descriptive caption $S = \{s_1, \ldots, s_n\}$, where each s_i is a sentence that effectively summarizes the content of the image. Consider $\mathcal{O} = \{o_1, o_2, \ldots, o_m\}$ as the set of visible geographic objects within I, where each o_j represents an object identified in the remote sensing image. The relationships among these objects can be denoted as $\mathcal{R}$, where $\mathcal{R} \subseteq \mathcal{O} \times \mathcal{O}$ represents a set of tuples (o_i, o_k), indicating a significant interaction or spatial relationship between object o_i and object o_k. An effective captioning model for RSIC should not only *identify the objects* in $\mathcal{O}$ but also *interpret the relationships* in $\mathcal{R}$.

B. Model Architecture

As shown in Fig. 1, the proposed RS-MoE model consists of four modules, including an image encoder, a vision–language model (VLM) encoder, an LLM block, and an MoE block.

1) Image Encoder: Given the remote sensing image I as defined earlier, we employ an image encoder f_I to extract its visual features F_I. Specifically, $F_I = f_I(I)$, where $F_I \in \mathbb{R}^{C\times W'\times H'}$, in which C is the number of feature channels and W' and H' represent the downsampled spatial dimensions. For this purpose, we adopt ViT-G/14 [39], a Vision Transformer with nearly two billion parameters, as the backbone of the image encoder. To prevent overfitting and ensure stable feature extraction, the parameters of f_I are frozen during the entire training process.

2) VLM Encoder: Inspired by InstructBLIP [18], our VLM encoder, denoted as f_{VLM}, is specifically designed to generate visual features aligned with the given instructions T. The process begins with a set of learnable query embeddings $Q_T \in \mathbb{R}^{L\times C'}$, where L is the number of queries and C' is the feature dimension. These queries, combined with the instructions T, are processed through the self-attention layers of the encoder to produce an intermediate instruction-aware feature set F_{SA}, defined as

$$F_{\text{SA}} = \text{SelfAttention}(Q_T, T). \tag{1}$$

Next, the feature F_{SA} is integrated with the visual features F_I extracted by the image encoder using cross-attention layers, resulting in instruction-aware visual features F_{CA}

$$F_{\text{CA}} = \text{CrossAttention}(F_{\text{SA}}, F_I). \tag{2}$$

This step dynamically aligns the visual features with the task-specific instructions, tailoring them to the input context. Finally, the refined features F_{CA} pass through an FFN and a fully connected layer to yield the final feature representation F_{VLM}

$$F_{\text{VLM}} = \text{FFN}(F_{\text{CA}}). \tag{3}$$

By incorporating these steps, the VLM encoder ensures that the extracted features are both adaptable and insightful, effectively aligning visual and instruction-specific information to support downstream tasks.

3) LLM Block: In the first training stage, the learned instruction-aware features F_{VLM}, along with the corresponding instructions T, serve as input prompts to the LLM block f_l. The LLM block generates a descriptive caption $S = \{s_1, \ldots, s_n\}$ that effectively summarizes the content of the remote sensing image I by identifying the objects in $\mathcal{O}$ and interpreting their relationships in $\mathcal{R}$. Formally, this process can be expressed as

$$S = f_l(F_{\text{VLM}}, T). \tag{4}$$

In the first training stage, the LLM block is fine-tuned to initialize the weights of the LLM models used in the MoE block. This preparatory step ensures that the models are capable of accurately capturing both object-level details and relationship dynamics, equipping the system to generate precise and coherent captions in the subsequent stages.

4) MoE Block: Similarly, in the second training stage, the features generated by the VLM encoder, F_{VLM}, are directly input into the MoE block to facilitate caption generation. Unlike traditional MoE modules, the MoE block comprises a novel instruction router R and N distinct LLMs, denoted as $\{f_{l1}, f_{l2}, \ldots, f_{lN}\}$. The instruction router R dynamically generates N detailed and adaptive prompts based on the input instruction T and the instruction-aware VLM features F_{VLM}

$$\{P_1, P_2, \ldots, P_N\} = R(T, F_{\text{VLM}}). \tag{5}$$

Here, R incorporates an FFN that projects the combined instruction T and visual features F_{VLM} into the embedding space for each subtask. This dynamic prompt generation ensures that each P_i is tailored to its respective subtask. These tailored prompts are then directed by the instruction router to their corresponding LLMs, enabling each LLM to focus on producing precise and contextually relevant outputs for its assigned subtask. In our experiment, the task is divided into three subtasks: thematic interpretation, object identification, and analysis of object relationships. Each LLM f_{li} focuses on one subtask, with its output expressed as

$$s_i = f_{li}(P_i, F_{\text{VLM}}), \quad i = 1, 2, 3. \tag{6}$$

The final caption S is obtained by aggregating the outputs of all LLMs

$$S = \text{Aggregate}(\{s_1, s_2, s_3\}). \tag{7}$$

By minimizing the loss of each LLM output against its respective subtask-specific ground truth, each LLM is guided to specialize in a narrow aspect of the task, reducing complexity and enhancing precision. This modular design, combined with dynamically adjusted prompts during training, ensures that the instruction router provides task-relevant guidance. Generalization errors are computed based on the subtask-specific ground truth, ensuring that the evaluation aligns with each subtask's intended learning objective. This approach prevents misleading assessments that could arise from evaluating outputs against an overly broad reference. Ultimately, this framework leverages

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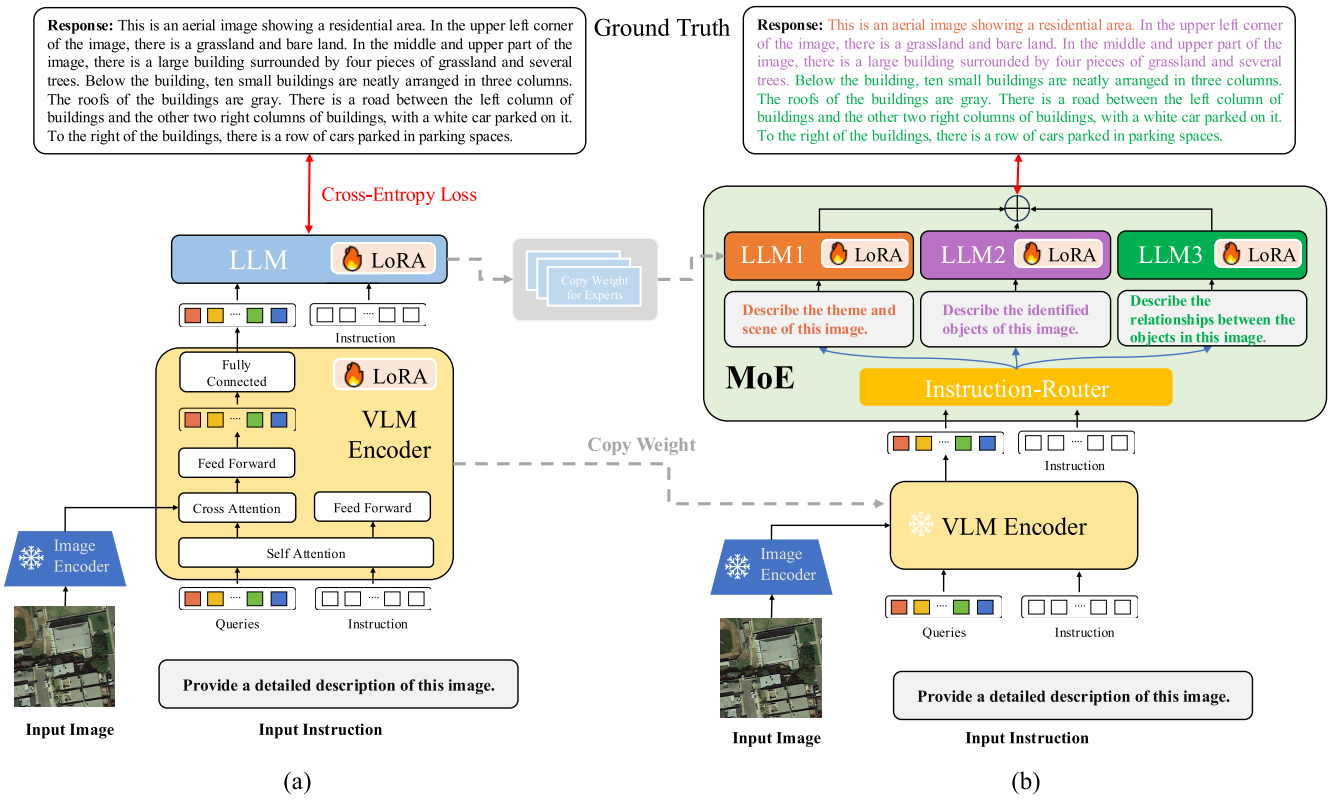


Fig. 1. Overview of the proposed RS-MoE model, which consists of four key components: the image encoder, the VLM encoder, the LLM block, and the MoE block. The MoE block comprises an instruction router that dynamically generates task-specific prompts and three lightweight LLMs, which focus on different aspects of the captioning task. In the generated captions, shown in the top-right corner of the figure, distinct colors represent each aspect: orange for the overall theme, purple for specific objects, and green for relationships between objects. RS-MoE is trained using a novel two-stage training strategy specifically designed for RSIC. (a) In Stage I, the VLM encoder and the LLM block are fine-tuned to initialize model weights specifically designed for the RSIC task. (b) In Stage II, the MoE block is fine-tuned to produce more detailed captions for RSIC tasks.

the combined expertise of multiple LLMs to significantly improve caption generation performance.

C. Model Tuning

As illustrated in Fig. 1, the tuning process of the proposed RS-MoE model is divided into two stages. Considering the complexity and diversity of information within remote sensing images, directly training our model can lead to performance degradation due to sparsity. This sparsity stems from the inadequate representation of remote sensing-specific features in models pretrained on natural images, creating inefficiencies in capturing the intricate characteristics of remote sensing imagery, ultimately impacting model performance. To address this, we propose a two-stage training strategy. The first stage begins by independently training the VLM encoder and LLM, ensuring that the encoder can extract rich and detailed visual features specific to remote sensing imagery while enhancing its capability to generate meaningful, context-aware feature representations. Once the VLM encoder generates stable visual features, the second stage involves fine-tuning the language models in the MoE. In this stage, a focus shifts to adjusting the MoE block's weights to produce more detailed captions tailored specifically for RSIC tasks, with each language model concentrating on its designated task.

1) Stage I: In this stage, our primary focus is on tuning the weights of the VLM encoder and the LLM block to initialize model weights specifically designed for the RSIC task. Specifically, we freeze the image encoder and then train the VLM encoder based on the extracted visual features and input instructions. This tuning enhances the encoder's ability to associate specific visual features with relevant instructional information, thereby improving its capability to generate meaningful and context-aware feature representations from remote sensing images. To significantly reduce the number of trainable parameters, we incorporate the LoRA training strategy during the VLM encoder training stage. First, within the self-attention layers, LoRA is specifically applied to the query and value projection layers. Second, LoRA is also implemented across all projection layers within the cross-attention layers, including those for the query, key, value, and output. Third, LoRA is integrated into the FFN components, ensuring a comprehensive application of this strategy throughout the model.

The LLM block receives output features from the VLM encoder and input instructions to generate the final captions for remote sensing images, which plays a crucial role in synthesizing the contextual data provided by the VLM encoder into coherent and detailed textual descriptions. The fine-tuning process is focused on enhancing the model's ability to understand

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the complex visual patterns typical of remote sensing images and to translate them into accurate linguistic representations. By tuning the LLM block, the model becomes more efficient at generating captions that are deeply aligned with the specific requirements of remote sensing tasks, including adapting to various landscapes, recognizing subtle environmental changes, and describing intricate relationships between different geographical features in the images. Similarly, we employ the LoRA technique to reduce the number of trainable parameters, thereby enhancing the model's efficiency and manageability during the training process. LoRA is specifically applied to the query and value layers within the attention module, as well as to the FFNs. This optimization not only streamlines the training process but also sharpens the focus of the attention mechanisms, improving the overall processing and responsiveness of the model.

2) Stage II: In this stage, our focus shifts to fine-tuning the weights of the LLMs within the MoE block to produce more detailed captions specifically for RSIC tasks. Tuning the MoE block involves detailed adjustments to the interaction mechanisms between the distinct LLMs to ensure that each LLM can effectively respond to its specific prompt by extracting and interpreting nuanced visual cues. This fine-tuning facilitates the development of a deep, segmented understanding of the image, allowing each expert within the MoE to generate a part of the caption that reflects its specialized knowledge of the image's theme, objects, or relationships. As these parts come together, they form a comprehensive and detailed caption that captures both the breadth and depth of the remote sensing imagery.

Specifically, we freeze the image encoder and the VLM encoder, where the weights of the VLM encoder are copied from those trained in Stage I. The MoE block uses the instruction router to convert the instruction into specific prompts, which are then combined with the features extracted by the VLM encoder related to the remote sensing images and input into different LLMs. By tuning the MoE, which principally involves tuning each of the sub-LLM blocks within it, we initialize the weights of each LLM block using the weights from the LLM blocks that were trained in Stage I. During the tuning process, we continue to employ the LoRA training strategy to further enhance the models' efficiency and effectiveness. This approach not only streamlines the training by reducing the number of parameters that need to be tuned but also assists in fine-tuning the model's ability to concentrate on specific features of the input data, thereby enhancing the overall accuracy and responsiveness of our system.

IV. Experiments

A. Datasets

1) RSICap: This dataset is built on the DOTA object detection dataset, which contains 2585 high-quality remote sensing image–text pairs for training VLMs on remote sensing images. The images are randomly chosen from the DOTA dataset and resized to 512 × 512 pixels. The text annotations are manually labeled by five experts in the remote sensing field. To evaluate the VLM model on this dataset, the authors further constructed an evaluation set, RSIEval, that consists of 1000 512 × 512 remote sensing images with manual annotations.

2) UCM-Captions: This dataset is derived from the UC Merced Land Use Dataset. It consists of 2100 remote sensing images, categorized into 21 land use classes, each containing 100 images. Each image measures 256 × 256 pixels, with a pixel resolution of 0.3048 m. The image captions were manually extracted from large-scale images provided by the United States Geological Survey's National Map Urban Area Imagery. Each image is described by five sentences, which are relatively fixed in structure and simple in their descriptions.

3) Sydney-Captions: This dataset is constructed from the Sydney dataset. It comprises 613 remote sensing images divided into seven land use categories. Each image measures 500 × 500 pixels, with a pixel resolution of 0.5 m. Each image is described in five sentences.

4) RSICD: This dataset contains 10 921 remote sensing images collected from four platforms: Google Earth, Baidu Maps, MapABC, and Tianditu. The images are uniformly resized to 224 × 224 pixels, although they come in various resolutions. The captions for these images were created by volunteers who possess experience in annotation and knowledge of remote sensing. For each image, the original caption varies from one to five sentences. The authors duplicated the captions that contained fewer than five sentences to ensure that each image had five distinct sentences in its caption.

5) VRSBench: This dataset consists of 29 614 images, each paired with a human-verified, detailed caption. These captions include abstract image attributes, such as source, resolution, and scene type, alongside object-specific details such as quantity, color, shape, and relative positioning. They emphasize clear, observable features, excluding ambiguous elements, and may describe additional objects, such as buildings, roads, and trees, beyond those in the source datasets. Each caption typically comprises 3–7 sentences, averaging 54 words.

B. Evaluation Metrics

Bilingual Evaluation Understudy (BLEU): It operates by comparing the text generated by the model to reference texts that are considered correct. BLEU examines the presence and frequency of n-grams in the generated text that appears in the reference text and calculates a score based on the precision of these n-grams. We tested BLEU scores for n-grams of sizes 1, 2, 3, and 4.

Recall-Oriented Understudy for Gisting Evaluation (ROUGE-L): It is a metric designed to evaluate the quality of summaries by measuring the longest common subsequence (LCS) between the generated text and the reference text. In the context of image captioning, ROUGE-L assesses the extent to which the content of a generated caption overlaps with that of a set of reference captions. This metric is recall-oriented, focusing on the coverage provided by the generated caption of the reference captions.

Metric for Evaluation of Translation With Explicit Ordering (METEOR): It is an evaluation metric that considers not only the precision but also the recall of matched words between the generated text and reference texts, thus balancing accuracy

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TABLE I
RESULTS OF OUR MODEL AND STATE-OF-THE-ART MODELS ON THE VRSBENCH DATASET FOR RSIC TASK

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
BLIP2-13B [16]	54.51	42.42	34.64	28.17	27.54	24.38	105.51
MiniGPT4-13B [17]	68.49	59.31	43.22	39.78	31.70	30.95	120.33
InstructBLIP-13B [18]	63.71	50.76	49.35	40.50	30.58	27.15	121.91
RSGPT-13B [20]	77.05	62.18	48.25	40.34	37.41	33.26	149.32
Qwen2-VL-7B [40]	78.66	60.98	47.07	39.24	40.05	32.63	124.36
LLaVA-NeXT-7B [41]	81.72	63.99	49.31	40.38	39.59	34.62	128.27
RS-MoE-1B	57.36	42.25	22.14	20.54	32.36	25.98	109.37
RS-MoE-3B	65.11	52.44	28.78	26.86	40.04	27.62	120.20
RS-MoE-7B	**82.13**	**65.44**	**51.93**	**42.55**	**40.28**	**35.72**	**158.36**

TABLE II
RESULTS OF OUR MODEL AND STATE-OF-THE-ART MODELS ON THE UCM-CAPTIONS DATASET FOR RSIC TASK

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
VLAD + RNN [44]	63.11	51.93	46.06	42.09	29.71	58.78	200.66
VLAD + LSTM [44]	70.16	60.85	54.96	50.30	34.64	65.20	231.31
mRNN [45]	60.10	50.70	32.80	20.80	19.30	-	214.00
mLSTM [45]	63.50	53.20	37.50	21.30	20.30	-	222.50
mGRU [46]	42.56	29.99	22.91	17.98	19.41	37.97	124.82
mGRU embedword [46]	75.74	69.83	64.51	59.98	36.85	66.74	279.24
CSMLF [47]	37.71	14.85	7.63	5.05	9.44	29.86	13.51
SAA [48]	79.62	74.01	69.09	64.77	38.59	69.42	294.51
Soft-attention [49]	74.54	65.45	58.55	52.50	38.86	72.37	261.24
Hard-attention [49]	81.57	73.12	67.02	61.82	42.63	76.98	299.47
SD-RSIC [50]	74.80	66.40	59.80	53.80	39.00	69.50	213.20
RTRMN (semantic) [51]	55.26	45.15	39.62	35.87	25.98	55.38	180.25
RTRMN (statistical) [51]	80.28	73.22	68.21	63.93	42.58	77.26	312.70
SVM-D BOW [52]	76.35	66.64	58.69	51.95	36.54	68.01	271.42
SVM-D CONC [52]	76.53	69.47	64.17	59.42	37.02	68.77	292.28
Post-processing [53]	79.73	72.98	67.44	62.62	40.80	74.06	309.64
RSGPT-13B [20]	86.12	79.14	72.31	65.74	42.21	78.34	333.23
SkyEyeGPT-7B [54]	90.71	85.69	**81.56**	**78.41**	46.24	79.49	236.75
RS-LLaVA-13B [55]	90.00	84.88	80.30	76.03	49.21	**85.78**	355.61
RS-CapRet-7B [56]	84.30	77.90	72.20	67.00	47.20	81.70	354.80
RS-MoE-7B w/o finetuning	**94.81**	**87.09**	79.57	72.34	**66.97**	62.74	**396.46**

and coverage. It also accounts for synonymy and stemming, allowing for a more nuanced comparison of content.

Consensus-Based Image Description Evaluation (CIDEr): It is specifically developed for scoring image captions by comparing them to reference captions written by humans. It measures the consensus between a candidate caption and the set of reference captions. This metric evaluates the similarity of n-grams between the generated and reference captions, weighting more frequent n-grams higher to reflect their importance in the descriptions.

C. Experimental Details

We trained our model on three NVIDIA A100 GPUs, each with 80 GB of memory. We selected three distinct versions of LLMs for use in our MoE model: Llama-3.2-1B [42], Llama-3.2-3B [42], and Vicuna-7B [43]. Each LLM within the MoE block was individually fine-tuned for only five epochs. During the training process, the learning rate was initially set to $1e^{-4}$, and a warm-up phase for the learning rate was applied for one epoch. We used the AdamW optimizer with parameters set to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 0.05, alongside adopting a cosine decay schedule for the reduction of the learning rate.

D. Results

1) Results on RSICap: We first evaluated the proposed RS-MoE model against state-of-the-art VLMs on the RSICap dataset for the RSIC task. Since these VLMs were initially designed for natural images, we fine-tuned each model according to the training strategies specified in their respective

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Ground Truth: This image is an aerial image of a residential area. There is a wide road in the middle of the image and another road on the right side. There are two rows of buildings above the image, each row with five buildings. There are seven swimming pools next to the buildings, one of which is empty. There are also two rows of buildings below the image, each row with five buildings, and eight swimming pools next to the buildings. There are seven cars in the image, three of which are driving on the road.

BLIP2: A satellite image of a city street with cars and buildings.

MiniGPT4: This is a satellite image of a neighborhood in the city of san diego, california. the neighborhood is made up of single-family homes with pools in the front yard. the houses are surrounded by trees and greenery. the streets are lined with palm trees and there

InstructBLIP: An image of a residential neighborhood with many houses lined up in a grid-like pattern. There are several swimming pools visible in the yards of the houses, as well as several cars parked on the streets. There is a road running through the middle of the neighborhood. The houses are surrounded by trees and shrubs, and there are some palm trees visible in the background.

RSGPT: This is an image of a residential neighborhood with many houses and swimming pools. There are a total of nine swimming pools visible in the image. The houses are well-maintained, and the swimming pools are clean and clearly visible. There are also several trees scattered throughout the neighborhood, and some of them are located near the swimming pools. A road runs through the center of the

RS-MoE: This is an image of a well-maintained residential neighborhood with many houses and swimming pools. There are many houses in the neighborhood, almost all of them have swimming pools. There are a total of twelve swimming pools visible in the image. In addition to the swimming pools, there are also several cars parked in the driveways of the houses. There is a road running through the middle of the neighborhood, and a power line runs around it. There are also several trees scattered throughout the neighborhood.



Ground Truth: This is an aerial image displaying an industrial area. A large white storage tank is located in the lower left corner of the image. Above it, there is a road and a white building. Ten small white storage tanks can be seen located above the road. Two white buildings are located in the upper left corner of the image. A wooded area can be seen on the right side of the image.

BLIP2: A satellite image of an industrial area with many trees.

MiniGPT4: The image shows a satellite view of a large industrial complex with several large tanks and other buildings. the tanks appear to be white and cylindrical in shape, with some smaller tanks.

InstructBLIP: This is a satellite image of an industrial area. There are several large white storage tanks in the center of the image, surrounded by trees and other buildings. There is a road running along the left side of the image, and a smaller road running along the right side. In the foreground, there is a wooded area.

RSGPT: This is a satellite image of an industrial area. There are some large white buildings. There is a road running along the bottom of the image, and a smaller road branching off to the left. There is also a small body of water visible in the top right corner of the image. Overall, the image depicts an industrial area with several large storage tanks and various other buildings.

RS-MoE: This is a satellite image of an industrial area. There are several large white storage tanks in the left corner of the image, surrounded by trees and other buildings. There is a road running along the left side of the image, and a smaller road running along the right side. There is a forest in the right side of the image. There are some storage tanks to the left of the forest. A road runs through the middle of the industrial area, above the road there are many small storage tanks, below the road is a large storage tank.



Ground Truth: This is an aerial image showing a suburban area. On the right side of the image, there is a grassland surrounded by trees. On the left side of the image, there are three buildings with two trucks and five cars below them. There is a curved road around the buildings. Below the image, there is a river flowing.

BLIP2: An aerial image shows an rural area.

MiniGPT4: The image shows a satellite view of farmland. There is farmland surrounded by many trees, and there are some houses next to the farm. There is a parking lot with some cars in it.

InstructBLIP: This image is an aerial view of a rural area with a river running through it. There are trees on either side of the river, and there are some buildings visible in the distance. In the foreground, there is a field with crops growing in it. There is a road running along the edge of the field, and there are some vehicles on the road. In the lower left corner of the image, there is a small parking lot with several cars parked in it. There is also a building visible in the lower left corner. Overall, the image shows a peaceful, rural landscape with a mix of natural and man-made features.

RSGPT: This is an aerial image of a rural area. There is a river running through the center of the image, with trees and shrubs growing along its banks. On the left side of the image, there is a road running parallel to the river. There are some buildings visible in the distance. In the foreground, there is a small parking lot with several cars parked in it. There are also some utility poles and power lines running along the road. Overall, the image shows a peaceful, rural landscape with a mix of natural and man-made features.

RS-MoE: This is an aerial image of a rural area with a river running through it. There is a lot of green vegetation in the area, including trees and grass. There is also a parking lot with several cars parked in it. In the uper right of the image, there is a field with some kind of crop growing in it. There is a road next to the river. There are several buildings on the left side of the road and there is a farm on the right side of the road. There is a farm surrounding by tree, and there is some crops in the farm.

Fig. 2. Qualitative results of image captions generated by our RS-MoE model and the SOTA model for three randomly selected remote sensing images from the RSEval dataset. In captions generated by our model, different colors indicate distinct aspects of the image: orange for the overall theme, purple for objects within the scene, and green for spatial relationships between objects.

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TABLE III
RESULTS OF OUR MODEL AND STATE-OF-THE-ART MODELS ON THE SYDNEY-CAPTIONS DATASET FOR RSIC TASK

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
VLAD + RNN [44]	56.58	45.14	38.07	32.79	26.72	52.71	93.72
VLAD + LSTM [44]	49.13	34.12	27.60	23.14	19.30	42.01	91.64
mRNN [45]	51.30	37.50	20.40	19.30	18.50	-	161.00
mLSTM [45]	54.60	39.50	22.30	21.20	20.50	-	186.00
mGRU [46]	69.64	60.92	52.39	44.21	31.12	59.17	171.55
mGRU embedword [46]	68.85	60.03	51.81	44.29	30.36	57.47	168.94
CSMLF [47]	59.98	45.83	38.69	34.33	24.75	50.18	75.55
SAA [48]	68.82	60.73	52.94	45.39	30.49	58.20	170.52
Soft-attention [49]	73.22	66.74	62.23	58.20	39.42	71.27	249.93
Hard-attention [49]	75.91	66.10	58.89	52.58	38.98	71.89	218.19
SD-RSIC [50]	72.40	62.10	53.20	45.10	34.20	63.60	139.50
SVM-D BOW [52]	77.87	68.35	60.23	53.05	37.97	69.92	227.22
SVM-D CONC [52]	75.47	67.11	59.70	53.08	36.43	67.46	222.22
Post-processing [53]	78.37	69.85	63.22	57.17	39.49	71.06	255.53
RSGPT-13B [20]	82.26	75.28	68.57	62.23	41.37	74.77	273.08
RS-CapRet-7B [56]	78.70	70.00	62.80	56.40	38.80	70.70	239.20
RS-MoE-7B w/o finetuning	**84.56**	**76.10**	**69.25**	**63.02**	**56.23**	**76.62**	**279.76**

papers to ensure optimal adaptation for remote sensing tasks. To ensure a fair comparison, all models were fine-tuned for five epochs, aligning with the training regime of our RS-MoE. Subsequently, we tested all models, including three versions of RS-MoE, on the RSIEval dataset to assess their captioning accuracy. The results, displayed in Table I, demonstrate that our RS-MoE-7B model outperformed other models across all seven evaluation metrics, including four BLEU score variants, as well as METEOR, ROUGE-L, and CIDEr scores.

Notably, our two lightweight variants, RS-MoE-1B and RS-MoE-3B, achieved performance comparable to the larger 7B and 13B state-of-the-art models and even outperformed them on certain metrics. For instance, RS-MoE-1B exceeded BLIP2-13B in BLEU-1, METEOR, ROUGE-L, and CIDEr scores. In addition, RS-MoE-1B not only surpassed BLIP2-13B but also outperformed MiniGPT4-13B and InstructBLIP-13B on the METEOR metric. These findings highlight the remarkable efficiency of our smaller models, as RS-MoE-1B and RS-MoE-3B deliver competitive performance while requiring significantly fewer parameters than their larger counterparts. Such compact architectures are particularly advantageous for resource-constrained environments, as they offer reduced computational overhead and are ideal for deployment in practical settings with limited memory and processing power.

Overall, the RS-MoE model demonstrates exceptional efficiency and accuracy in addressing the RSIC task. By leveraging its innovative architecture, including the MoE framework and specialized training strategy, the model effectively captures and interprets the complex visual information inherent in remote sensing images. This allows for the generation of precise, contextually relevant captions. Thus, our model not only excels in terms of performance but also proves highly practical for real-world applications, especially in environments where computational resources are limited.

To more intuitively compare the captioning performance of our model with other models for given remote sensing images, we randomly selected three images from the RSIEval dataset and listed the captions generated by each model in Fig. 2. These three images represent three distinct scenes: urban streets, industrial areas, and rural regions. By comparing the results of the generated captions from different models, several key findings can be made. The BLIP2 model provides relatively simple image captions, focusing mainly on the primary objects in the remote sensing images. The MiniGPT4 and InstructBLIP models incorporate more details and contextual information in their descriptions, such as trees around houses, swimming pools, and roads. These details enrich the descriptions but can sometimes result in information redundancy. The RSGPT model describes the images in a more structured manner. However, some of its generated descriptions may lack accuracy, such as the number of swimming pools, and may include some subjective and hallucinatory sentences. Conversely, our RS-MoE model describes the images with clear logic from three perspectives: themes, objects present, and the relationships between objects. The RS-MoE model provides more accurate descriptions of the types, quantities, and spatial relationships of the objects present in images.

2) Results on UCM-Captions: We further validated the performance of our model on the classic UCM-Captions dataset. To establish a comprehensive comparison, we selected 16 deep learning models specifically designed for the RSIC task, representing state-of-the-art approaches in RSIC prior to the advent of VLM models. In addition, we evaluated recent state-of-the-art VLM models in the remote sensing domain, including RSGPT [20], SkyEyeGPT [54], RS-LLaVA [55], and RS-CapRet [56]. Notably, we evaluated the performance of the 7B version of RS-MoE, which was trained on the RSICap dataset *without any additional fine-tuning* on the

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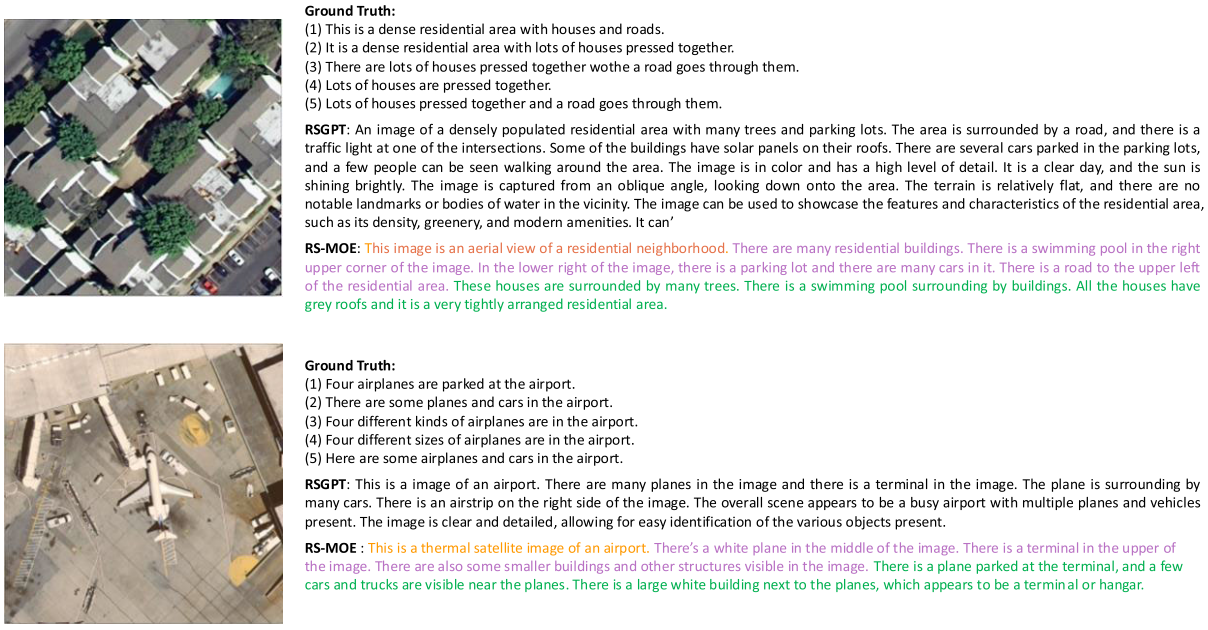


Fig. 3. Qualitative results of image captions generated by RSGPT and our RS-MoE model for two randomly selected remote sensing images from the UCM-Captions dataset. In captions generated by our model, different colors indicate distinct aspects of the image: orange for the overall theme, purple for objects within the scene, and green for spatial relationships between objects.

UCM-Captions dataset. The comparative results of these baseline models and our model are presented in Table II. Remarkably, even without fine-tuning on the UCM-Captions dataset, our model achieved the highest performance across several metrics, including BLEU-1, BLEU-2, METEOR, and CIDEr. Specifically, our model achieved the METEOR score of 66.97, significantly surpassing the second-highest score of 49.21 obtained by the RS-LLaVA-13B model, further demonstrating the robustness and generalizability of RS-MoE across different datasets.

To further illustrate the effectiveness of our model in generating accurate and contextually aware captions, we selected two remote sensing images with distinct scenes from the test set of the UCM-Captions dataset. The captions generated by RSGPT and our model for these images are shown in Fig. 3. As observed, both models successfully capture the overall themes of the images. However, for detailed descriptions, particularly in the first remote sensing image, the captions generated by RSGPT exhibit discrepancies with the actual scene. In contrast, the captions produced by the RS-MoE model provide precise and detailed representations of both the objects and their spatial relationships within the remote sensing images, underscoring our model's advantage in understanding complex spatial arrangements in diverse remote sensing imagery.

3) Results on Sydney-Captions: To further assess our model's efficacy for the RSIC task, we extended the evaluation to the traditional Sydney-Captions dataset. In this experiment, 16 deep learning models specifically designed for RSIC, along with two recent VLM models, RSGPT [20] and RS-CapRet [56], were selected as comparative benchmarks. Note that we still employed the 7B variant of RS-MoE *without any additional fine-tuning* on the Sydney-Captions dataset. The comparative performance of these baseline models and our model is detailed in Table III. Analysis of the results demonstrates that even without fine-tuning, our model consistently outperformed all competing models across all seven evaluation metrics. A particularly notable achievement is its performance on the METEOR metric, where it achieved a score of 56.23, surpassing the next best model's score of 41.37 by a significant margin of 15%.

To further illustrate the captioning capabilities of our model in comparison to RSGPT, we randomly selected two remote sensing images from the test set of the Sydney-Captions dataset, as shown in Fig. 4. These images depict scenes of a residential area and an airport. Fig. 4 shows that while both RSGPT and RS-MoE provide generally accurate descriptions of the overall themes and specific details, the RSGPT-generated captions lack the depth observed in our model's outputs. In addition, the last sentence generated by RSGPT for the first image remains incomplete, underscoring our model's advantage in generating contextual descriptions.

4) Results on RSICD: Finally, we validated the performance of our model on the widely used RSICD dataset for the RSIC task. Consistent with previous evaluations, we compared our model against 16 deep learning models and two recent VLM models, RSGPT [20] and RS-CapRet [56], with our RS-MoE-7B model evaluated *without additional fine-tuning* on the RSICD dataset. The comparative results for these baseline models and our model are presented in Table IV. As observed in Table IV, even without fine-tuning, our model achieved the highest scores on five out of seven metrics, including BLEU-1, BLEU-2, BLEU-3, BLEU-4, and METEOR. This highlights the robustness and adaptability of our approach, underscoring its ability to perform well across diverse datasets without requiring dataset-specific adjustments.

To further illustrate the effectiveness of the generated captions, we selected two remote sensing images from the test

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TABLE IV
RESULTS OF OUR MODEL AND STATE-OF-THE-ART MODELS ON THE RSICD DATASET FOR RSIC TASK

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
VLAD + RNN [44]	49.38	30.91	22.09	16.77	19.96	42.42	103.92
VLAD + LSTM [44]	50.04	31.95	23.19	17.78	20.46	43.34	118.01
mRNN [45]	45.58	28.25	18.09	12.13	15.69	31.26	19.15
mLSTM [45]	50.57	32.42	23.29	17.46	17.84	35.02	31.61
mGRU [46]	42.56	29.99	22.91	17.98	19.41	37.97	124.82
mGRU embedword [46]	60.94	46.24	36.80	29.81	26.14	48.20	159.54
CSMLF [47]	57.59	38.59	28.32	22.17	21.28	44.55	52.97
SAA [48]	59.35	45.11	35.29	28.08	26.11	49.57	132.35
Soft-attention [49]	65.13	49.04	39.00	32.30	26.39	49.69	90.58
SD-RSIC [50]	64.50	47.10	36.40	29.40	24.90	51.90	77.50
RTRMN (semantic) [51]	62.01	46.23	36.44	29.71	28.29	55.39	151.46
RTRMN (statistical) [51]	61.02	45.14	35.35	28.59	27.51	54.52	148.20
SVM-D BOW [52]	61.12	42.77	31.53	24.11	23.03	45.88	68.25
SVM-D CONC [52]	59.99	43.47	33.55	26.89	22.99	45.57	68.54
MLAT [57]	66.90	51.13	41.14	34.21	27.31	50.57	94.27
Post-processing [53]	62.90	45.99	35.68	28.68	25.30	47.34	75.56
RSGPT-13B [20]	70.32	54.23	44.02	36.83	30.10	53.34	102.94
RS-CapRet-7B [56]	72.00	59.90	50.60	43.30	37.00	**63.30**	**250.20**
RS-MoE-7B w/o finetuning	**78.61**	**63.60**	**51.44**	**43.35**	**39.30**	60.77	109.46



Ground Truth:
(1) Lots of houses with different colors of roofs arranged neatly.
(2) A residential area with houses arranged neatly and some roads go through this area.
(3) There are many houses arranged neatly in the residential area while some roads go through.
(4) There is a residential area with houses arranged neatly while many plants on the roadside.
(5) A residential area with houses densely arranged.

RSGPT: This is a satellite image of a densely populated urban area. There are many high-rise residential buildings, roads, and trees visible in the image. The area appears to be an inner-city suburb with a mix of houses and apartments. There are also several parks and green spaces visible in the image. The image also shows the mix of old and new buildings in the area, with some buildings appearing to be quite old while others are more modern. There are also several businesses and shops visible in the

RS-MoE: This is a satellite image of a residential area of the city. There are three large residential areas in the image. There are many buildings in the image. There are also several trees scattered throughout the neighborhood. In the middle of the image there is a car park with many cars parked in it. There are two roads on either side of the image that run through the entire residential area. There is a lot of cars are traveling. In the bottom left corner of the image residential are some white flats and at the bottom of the image there is a small park. This is a very busy residential area.



Ground Truth:
(1) Three airplanes parked on the airport with some luggage cars.
(2) Three white airplanes parked neatly on the airport with an airport building beside.
(3) There are three same airplanes parked neatly on the airport with some luggage cars beside.
(4) Three airplanes parked on the airport with some luggage cars beside.
(5) An airport with three white airplanes parked on while some luggage cars beside.

RSGPT: The image is an aerial view of an airport, showing several airplanes parked on the tarmac. There is a large white building in the center of the airport with several smaller planes parked around it. There is also a taxiway and a runway visible in the image. The image has a high level of detail and is labeled with the names of the different features visible in the scene.

RS-MoE: This image is an aerial view of an airport. There is a large building in the lower right corner of the image. To the left of the image is an airstrip and a grassy area. There is a lawn in the left side of the image. There are many planes surrounding the building. There is an airstrip in the upper right corner of the airport. There are also some vehicles parked inside the airport. All airplane and vehicles are surrounding the airport.

Fig. 4. Qualitative results of image captions generated by RSGPT and our RS-MoE model for two randomly selected remote sensing images from the Sydney-Captions dataset. In captions generated by our model, different colors indicate distinct aspects of the image: orange for the overall theme, purple for objects within the scene, and green for spatial relationships between objects.

set of the RSICD dataset and displayed the captioning results generated by RSGPT and our model in Fig. 5. It is evident from Fig. 5 that, unlike RSGPT, our RS-MoE model accurately describes the spatial positions of objects within the images. This capability demonstrates RS-MoE's enhanced ability to capture and articulate spatial relationships and arrangements, which are essential for a comprehensive understanding of remote sensing imagery. By providing detailed, contextually rich descriptions that reflect the actual scene, RS-MoE ensures that the generated captions are both informative and precise, thereby improving the overall interpretability.

5) Results on VRSBench: We further evaluate our method on the RSBench dataset, a benchmark specifically designed to test detailed captioning capabilities. In our experiments, we compare our method against several strong baselines, including GeoChat without finetuning [21], GPT-4V [58], MiniGPT-v2 [59], LLaVA-1.5 [60], GeoChat [21], and Mini-Gemini [61]. The results of these models are summarized in

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Ground Truth:
(1) some green trees are near a large piece of bareland.
(2) some planes are parked near an airport with parking lot.
(3) some planes are parked near an airport with parking lot.
(4) some green trees are near a large piece of bareland.
(5) some green trees are near a large piece of bareland.

RSGPT: This is a satellite image of an airport. There are several buildings and parking lots visible, as well as a runway and taxiways. There are also some trees and a body of water near the airport. The surrounding area appears to be mostly flat and covered in grass or other vegetation. The image has been enhanced to show more detail.

RS-MoE: This image is a low-resolution aerial photograph of an airport. There are a lot of trees in the lower left corner of the image. There's an airport in the right of the image, with many planes and two large write buildings. There is a forest in the upper right of the image. There is an airport in the right of the image , to the right of the airport are two white buildings and to the right are some aeroplanes. On the left side of the airport is a forest and some grey buildings, and there are many roads through the airport.



Ground Truth:
(1) a highway is built over the river.
(2) the land along the river is full of grass.
(3) a curved bridge is built above the sea and land.
(4) a curved bridge is built above the sea and land.
(5) a highway is built over the river.

RSGPT: The image is an aerial view of a river with several bridges crossing over it. There are several artificial islands in the middle of the river. The area around the river is mostly green grass and trees. There are some buildings visible in the distance. There are some ships on the river. The image is clear and detailed, showing the various structures and landscapes in the area.

RS-MoE: This is a satellite overlook of a river. In the centre of the image there is a man-made island that divides the river into two parts. There is a waterfall on the left side of the image. There are some trees on both sides of the river. There is a bridge that crosses the river over the island. There are some cars on the bridge. On the left of the island there is a waterfall .

Fig. 5. Qualitative results of image captions generated by RSGPT and our RS-MoE model for two randomly selected remote sensing images from the RSICD dataset. In captions generated by our model, different colors indicate distinct aspects of the image: orange for the overall theme, purple for objects within the scene, and green for spatial relationships between objects.

TABLE V
RESULTS OF OUR MODEL AND STATE-OF-THE-ART MODELS ON THE VRSBENCH DATASET FOR RSIC TASK

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	CHAIR
GeoChat w/o ft [21]	13.9	6.6	3.0	1.4	7.8	13.2	0.4	0.42
GPT-4V [58]	37.2	22.5	13.7	8.6	20.9	30.1	19.1	**0.83**
MiniGPT-v2 [59]	36.8	22.4	13.9	8.7	17.1	30.8	21.4	0.73
LLaVA-1.5 [60]	48.1	31.5	21.2	14.7	21.9	36.9	33.9	0.78
GeoChat [21]	46.7	30.2	20.1	13.8	21.1	35.2	28.2	0.77
Mini-Gemini [61]	47.6	31.1	20.9	14.3	21.5	36.8	33.5	0.77
RS-MoE	**55.3**	**34.9**	**25.5**	**17.2**	**25.1**	**42.8**	**40.5**	0.81

Table V. As shown in the table, our method demonstrates outstanding performance across all major metrics. Notably, our method achieves a BLEU-1 score of 55.3, significantly surpassing Mini-Gemini's BLEU-1 score of 47.6. Similarly, our method achieves a BLEU-3 score of 25.5, outperforming Mini-Gemini's 20.9. Although GPT-4V achieves the highest CHAIR score of 0.83, our method achieves a competitive score of 0.81, substantially outperforming other baselines. These results clearly demonstrate the effectiveness of our method in generating detailed and descriptive captions that capture both abstract image attributes and object-specific details. The substantial improvements over state-of-the-art baselines on RSBench further highlight the robustness and adaptability of our method in addressing the unique challenges of vision–language modeling for detailed captioning tasks.

V. DISCUSSION

In this section, we conduct a comprehensive analysis of the effectiveness and generalizability of the proposed RS-MoE model from two perspectives. First, we verify the effectiveness of our model and training strategy through four ablation studies. These include examining the effect of the instruction router, varying the number of LLMs in the MoE block, using different types of LLMs, and evaluating the two-stage training strategy. Subsequently, we extend the application of our model to the RSVQA task, testing its performance on two datasets, RSEval and RSIVQA, to assess its broader applicability and generalization capabilities.

A. Ablation Study

1) Effect of Instruction Router: To evaluate the effectiveness of the instruction router within the MoE block, we conducted an ablation study comparing the model's performance with and without the instruction router. For a fair comparison, both setups use three LLMs as expert models. When the instruction router is absent, the input instructions are directly passed to the LLMs without any dynamic prompt generation, and each LLM independently generates outputs. The results of the two models are shown in Table VI. It can be found that the model without the instruction router achieves the BLEU-1 score of 47.32, the METEOR score of 22.87, and the CIDEr score of 102.51. In contrast, incorporating the

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TABLE VI
COMPARISON RESULTS OF RS-MOE WITH AND WITHOUT THE INSTRUCTION ROUTER ON THE RSICAP DATASET FOR THE RSIC TASK

Training Strategy	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
RS-MoE w/o Instruction Router	47.32	33.18	25.66	21.42	22.87	25.33	102.51
RS-MoE w/ Instruction Router	82.13	65.44	51.93	42.55	40.28	35.72	158.36

TABLE VII
COMPARISON RESULTS OF DIFFERENT NUMBERS OF LLMS IN THE MOE BLOCK ON THE RSICAP DATASET FOR THE RSIC TASK

Number of LLMs	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
1	48.51	38.65	29.57	25.13	23.01	21.49	93.51
2	64.52	51.32	40.43	33.42	30.61	28.27	124.41
3	**82.13**	**65.44**	**51.93**	**42.55**	**40.28**	**35.72**	**158.36**
4	77.36	60.38	48.91	39.26	37.94	32.95	147.24

instruction router leads to significant improvements, with the model achieving the BLEU-1 score of 82.13, the METEOR score of 40.28, and the CIDEr score of 158.36. These results highlight the critical role of the instruction router in dynamically generating tailored prompts based on input instructions and visual features, effectively guiding the expert models to produce more precise and contextually appropriate descriptions. The substantial performance gains observed with the instruction router emphasize its importance in enabling the model to adapt to diverse input images and tasks. By leveraging prompt learning principles, the instruction router enhances the model's ability to extract meaningful visual information and generate accurate, detailed descriptions, demonstrating its indispensability in the proposed framework.

2) Effect of Different Numbers of LLMs in the MoE Block: As described in Section III-B, within the MoE block, we employ an instruction router to generate three detailed prompts, each directed to a distinct LLM to independently handle the overall theme, geographic objects, and relationships among objects in remote sensing images. In this section, we analyze the results obtained from using varying numbers of LLMs. Specifically, we trained versions of the RS-MoE model with one to four LLMs on the RSICap dataset, keeping the first stage constant while adjusting the number of LLMs in the second stage. When the number of LLMs is set to 1, the model reduces to a conventional single-LLM decoder VLM. With two LLMs, each LLM is tasked with describing either the image's theme or specific information. For configurations with four LLMs, we assign one LLM each to address the image's theme, objects, absolute positions of objects, and relationships among objects.

We evaluated the performance of these models on the RSIC task using the RSIEval dataset, with detailed results presented in Table VII. By comparing the results for configurations with more than one LLM to those with only a single LLM, we observe substantial performance improvements when the task is divided into subtasks. These results further validate the effectiveness of incorporating the idea of MoE into the RSIC task. In addition, comparing the results for models with two, three, and four LLMs, we found that using three LLMs yields the best performance. Consequently, in our experiments, we employ three LLMs to focus on generating captions for the image's theme, included objects, and spatial relationships among objects, highlighting the advantage of the MoE framework for handling complex, multifaceted tasks in remote sensing. This setup not only enhances the interpretability of the generated captions but also demonstrates the flexibility and scalability of our model when addressing intricate details in remote sensing imagery.

3) Effect of Different LLM Models: In this section, we investigate the effect of different lightweight LLMs within the MoE block on RSIC. Specifically, in addition to the original selection of Llama-3.2-1B, Llama-3.2-3B, and Vicuna-7B, we further experimented with four additional state-of-the-art LLMs from the computer vision field: Qwen-1.8B [63], StableLM-1.6B [62], Phi2-2.7B [64], and LLama-7B [65]. These models are designed to understand and generate human-like text across a wide range of topics, and we selected them to assess how different LLM configurations influence performance within our proposed RS-MoE.

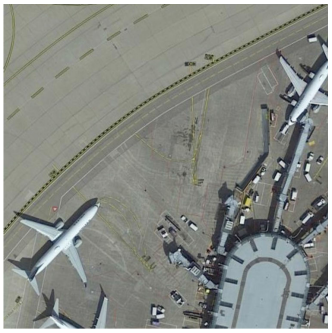
Each of the seven RS-MoEmodel variants was trained as a baseline on the RSICap dataset and evaluated on the RSIEval dataset. The performance metrics are displayed in Table VIII. As observed from the results, all selected lightweight LLMs under our MoE framework achieved reasonably satisfactory performance in RSIC. This consistency across different LLMs demonstrates the robustness and adaptability of our proposed framework, indicating that even with various lightweight LLM configurations, the RS-MoEmodel is able to maintain effective and contextually relevant caption generation for complex remote sensing imagery. This suggests that our MoE framework can be flexibly integrated with different LLMs, providing a versatile solution adaptable to the needs of various tasks and computational environments.

4) Effect of Two-Stage Training Strategy: We further evaluate the effectiveness of our proposed two-stage training strategy. As explained in Section III-C, directly training our RS-MoEmodel can lead to performance degradation due to sparsity issues stemming from the distinct characteristics of remote sensing images compared to natural images.

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TABLE VIII
COMPARISON RESULTS OF DIFFERENT LLM MODELS IN THE MOE BLOCK ON THE RSICAP DATASET FOR THE RSIC TASK

LLM in MoE Block	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
Llama-3.2-1B [42]	57.36	42.25	22.14	20.54	32.36	25.98	109.37
StableLM-1.6B [62]	56.42	44.36	23.27	22.14	33.71	24.62	110.85
Qwen-1.8B [63]	59.35	47.28	27.52	22.95	34.62	25.38	113.98
Phi2-2.7B [64]	59.78	48.15	27.35	25.58	36.77	25.36	114.96
Llama-3.2-3B [42]	65.11	52.44	28.78	26.86	40.04	27.62	120.20
LLaMa-7B [65]	77.95	62.11	49.28	40.27	38.26	31.91	150.31
Vicuna-7B[43]	**82.13**	**65.44**	**51.93**	**42.55**	**40.28**	**35.72**	**158.36**



Question: What is the theme of the image? (***Theme***)

Ground truth: Airport.

BLIP2:	This is an image of airplanes.
InstructBLIP:	An airport.
MiniGPT4:	The theme of the image is an airport.
RSGPT:	The theme of the image is an airport.
RS-MoE:	The theme of the image is an airport.

Question: How many airplanes are there in this image? (***Quantity***)

Ground truth: Three.

BLIP2:	There are two airplanes in this image.
InstructBLIP:	There are three airplanes in this image.
MiniGPT4:	There are two airplanes parked at the gate in this image.
RSGPT:	There are two airplanes in this image.
RS-MoE:	There is an airport in the image with three airplanes in it.

Question: Where is the terminal located in the image? (***Absolute Position***)

Ground truth: Lower left conner.

BLIP2:	There is a terminal in the image.
InstructBLIP:	The terminal is in the upper right corner of the image.
MiniGPT4:	The terminal located in the left of the image.
RSGPT:	The terminal is in the lower part of the image.
RS-MoE:	There is a terminal in the lower left corner of the image.

Question: What is the relative position between the terminal and the airplanes? (***Relative Position***)

Ground truth: Surrounding the terminal.

BLIP2:	The airplanes is located in the lower left corner.
InstructBLIP:	The terminal is closer to the airplanes than they are to each other.
MiniGPT4:	The relative position between the terminal and the airplanes is that the airplanes are parked on the tarmac in front of the terminal building.
RSGPT:	The airplanes is above the terminal.
RS-MoE:	There are some airplanes surrounding the terminal.

Fig. 6. Qualitative results of VQA for remote sensing images by BLIP2, MiniGPT4, InstructBLIP, RSGPT, and our RS-MoE model on the RSVQA test set from the RSEval dataset.

To validate this hypothesis, we implemented a baseline model trained using a single-stage approach, directly optimizing the model without employing the two-stage strategy. Comparative results for both training methods are shown in Table IX. The

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TABLE IX
COMPARISON RESULTS OF DIFFERENT TRAINING STRATEGY FOR OUR MODEL ON THE RSICAP DATASET FOR THE RSIC TASK

Training Strategy	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
One-Stage	50.39	33.44	24.37	20.86	27.49	25.36	98.21
Two-Stage	82.13	65.44	51.93	42.55	40.28	35.72	158.36

TABLE X
RESULTS OF OUR MODEL AND STATE-OF-THE-ART MODELS ON THE RSVQA TEST SET OF RSIEVAL FOR RSVQA TASK

Method	Presence	Quantity	Color	Absolute pos.	Relative pos.	Area comp.	Road dir.	Image	Scene	Reasoning	Avg accuracy
BLIP2	60.41	26.02	43.24	7.69	13.16	58.14	33.33	74.42	43.24	47.50	45.56
MiniGPT4	29.70	9.76	31.53	1.54	1.32	16.28	0.00	34.88	24.32	17.50	21.82
InstructBLIP	76.14	21.95	45.05	12.31	10.53	69.77	0.00	81.40	45.95	57.50	53.26
RSGPT	81.22	**39.02**	54.05	38.46	35.53	62.79	**66.67**	**93.02**	89.19	70.00	65.24
RS-MoE	**83.16**	32.77	**71.92**	**43.95**	**45.00**	**63.41**	0.00	77.89	**93.18**	**74.07**	**68.65**

TABLE XI
RESULTS OF OUR MODEL AND STATE-OF-THE-ART MODELS ON THE RSIVQA DATASET FOR RSVQA TASK

Model	Yes/No	Number	Others	Average Accuracy	Overall Accuracy
MAIN [66]	92.82	56.71	54.50	68.01	77.39
EasyToHard [67]	95.49	49.03	63.65	69.39	79.70
SHRNet [68]	97.64	57.89	84.60	80.04	84.46
RSAdapter [69]	97.90	62.64	92.47	84.34	87.10
RS-MoE	**98.01**	**75.46**	**92.88**	**88.78**	**89.12**

findings indicate that the two-stage training strategy significantly outperforms the single-stage approach, underscoring the advantages of this method. By incorporating the first training stage, the VLM encoder and LLM block learn feature representations tailored to remote sensing imagery, gaining a foundational understanding of remote sensing-specific visual patterns and linguistic cues. This process establishes a robust basis for accurate feature extraction and preliminary captioning. This stage essentially fine-tunes the model's ability to interpret complex spatial arrangements and thematic elements unique to remote sensing imagery, elements often underrepresented in models pretrained on natural images. In the second stage, the model builds upon this solid foundation, requiring only minimal fine-tuning to adapt the MoE block. Consequently, each LLM in the MoE block can effectively focus on its designated task, whether it involves capturing thematic elements, identifying specific geographic objects, or interpreting spatial relationships among objects. This task-oriented fine-tuning results in captions that are not only precise but also provide a comprehensive view of the remote sensing scene, with detailed and contextually relevant descriptions. Overall, this two-stage training strategy represents a core innovation of our work, demonstrating how carefully designed training frameworks can achieve superior results in the specialized domain of remote sensing.

B. Extension on the RSVQA Task

To verify the generalization of the proposed model, we further evaluated its performance on the visual question answering (VQA) task for remote sensing images. Specifically, we used the RSEval dataset's ten categories of questions, including presence, quantity, color, absolute position, relative position, area comparison, road direction, image, scene, and reasoning. We queried BLIP2, MiniGPT4, InstructBLIP, RSGPT, and our RS-MoE model, all of which were trained on the RSICap dataset, and calculated the average accuracy of their answers for each category. As shown in Table X, our model achieved the highest performance in seven out of the ten categories. To visually demonstrate the results on RSVQA, we randomly selected a remote sensing image from the dataset and displayed the answers provided by these models in four aspects: theme, quantity, absolute position, and relative position. It was observed from Fig. 6 that all models were able to answer simple theme questions accurately. However, as the complexity of the questions increased, especially for those related to spatial relationships, the baseline models' performance deteriorated, while our model continued to provide accurate answers. This indicates that our model excels in handling complex questions, demonstrating its superior capability in understanding and reasoning about spatial relationships in remote sensing images.

To evaluate the effectiveness of our model further, we conducted experiments on the RSIVQA dataset, a benchmark for the VQA task in remote sensing. The RSIVQA dataset includes three question categories: yes/no, number, and others, with the final results summarized in Table XI. Our model is compared against state-of-the-art methods, including MAIN [66], EasyToHard [67], SHRNet [68], and

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RSAdapter [69]. As shown in the table, our model achieves the best performance across all metrics, with an overall accuracy of 89.12% and an average accuracy of 88.78%. Notably, for the challenging number category, our model achieves a significant improvement, with an accuracy of 75.46%, outperforming the previous best model, RSAdapter, by 12.82%. Similarly, in the yes/no and others categories, our model attains accuracies of 98.01% and 92.88%, respectively, setting new benchmarks in both cases. These results highlight the superior generalization ability of our model in handling diverse and complex question types in the RSIVQA dataset. Its exceptional performance, particularly in the number and others categories, underscores its capability to reason about numerical quantities and contextual relationships in remote sensing imagery. This demonstrates the robustness and adaptability of our approach in addressing the unique challenges posed by VQA tasks in remote sensing.

VI. Conclusion

This article introduces RS-MoE, a pioneering MoE-based VLM specifically developed for RSIC. RS-MoE consists of four core modules: the image encoder, VLM encoder, LLM block, and MoE block. At the heart of the framework, the MoE block leverages an instruction router to dynamically generate task-specific prompts, guiding each LLM in subsequent layers to focus on distinct aspects of captioning—theme comprehension, object identification, and relationship interpretation. This design allows the model to capture intricate semantic features unique to remote sensing imagery, enabling the generation of detailed, contextually accurate captions. In addition, we propose a two-stage training strategy to address performance degradation from sparsity issues inherent in MoE-based methods and enhance training efficiency. Our extensive experiments show that even after fine-tuning on only one dataset, RS-MoE achieves state-of-the-art performance across five RSIC datasets and demonstrates robust generalization capabilities on two RSVQA datasets. Notably, our lightweight RS-MoE-1B variant achieves performance comparable to larger 13B VLMs, underscoring the model's efficiency in remote sensing applications. Thus, RS-MoE not only excels in performance but also proves practical for real-world applications, particularly in resource-constrained environments where computational efficiency is essential.

While RS-MoE demonstrates significant advantages, there are certain limitations that warrant further exploration. The reliance on predefined subtasks for LLM specialization may limit adaptability to highly complex or novel remote sensing scenarios. In addition, the two-stage training strategy, while effective, introduces additional computational overhead during the pretraining phase. Future work could explore adaptive task routing mechanisms that allow for greater flexibility and generalization to new tasks. Furthermore, integrating multimodal data sources, such as radar and hyperspectral imagery, could expand the applicability of RS-MoE to a broader range of remote sensing challenges.

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